# Julian Asilis

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U.S. Citizen

## Education

08/2022 – 12/2026 **Ph.D. in Computer Science**, *University of Southern California*, Los Angeles
Supported by NSF Graduate Research Fellowship, 2024–present
*Research*: Statistical learning theory, deep generative models.
*Advisor*: Vatsal Sharan

2016 – 2020 **A.B. in Mathematics with High Honors**, *Harvard University*, Cambridge, MA
*Senior Thesis*: Probability Monads, under Michael Hopkins.

## Experience

Summer 2025 **Machine Learning Engineer Intern**, *Pinterest*, Seattle, WA
- Accelerated training of large-scale ads ranking model by 32% while improving AUC.
- Implemented data subsampling/reweighting methods, including local case-control (LCC) and reweighted SGD, to improve efficiency and reduce training costs.

2021 – 2022 **Research Associate**, *Boston College*, Chestnut Hill, MA
- Researched computable learning theory and topological measures of complexity for neural nets, leading to a publication at LICS and extended abstract at CCA.
- Served as TA for 2 computer science courses, including writing 140 pages of notes, overseeing 7 TA's, and writing scripts for automated exam grading.

2020 – 2021 **Quantitative Researcher**, *AQR Capital Management*, Greenwich, CT
- Refined and expanded several factors used to trade dozens of assets in fixed income.
- Performed statistical inference and time series modeling on large panel data.
- Wrote production code in Python and SQL.

## Research

Theory papers typically use alphabetical author ordering.

13. **Does Random Search Discover Task Experts? Revisiting Neural Thickets**
Julian Asilis, Guanzhe Hong, Vatsal Sharan.
In submission, 2026.

12. **Resa: Transparent Reasoning Models via SAEs**
Shangshang Wang, Julian Asilis, Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, Deqing Fu, Willie Neiswanger.
In *Conference on Language Modeling (COLM)*, 2026.

11. **Textual Steering Vectors Can Improve Visual Understanding in Multimodal Large Language Models**
Woody Gan, Deqing Fu, Julian Asilis, Ollie Liu, Dani Yogatama, Vatsal Sharan, Robin Jia, Willie Neiswanger.
In *Association for Computational Linguistics (ACL)*, 2026.

10. **Tina: Tiny Reasoning Models via LoRA**
Shangshang Wang, Julian Asilis, Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, Willie Neiswanger.
In *International Conference on Learning Representations (ICLR)*, 2026.

9. **Semi-Random Graphs, Robust Asymmetry, and Reconstruction**
Julian Asilis, Xi Chen, Dutch Hansen, Shang-Hua Teng.
In *Innovations in Theoretical Computer Science (ITCS)*, 2026.

8. **On Agnostic PAC Learning in the Small Error Regime**
Julian Asilis, Mikael Møller Høgsgaard, Grigoris Velegkas.
In *Neural Information Processing Systems (NeurIPS)*, 2025. **Spotlight paper**

7. **Local Regularizers Are Not Transductive Learners**
Sky Jafar, Julian Asilis, Shaddin Dughmi.
In *Conference on Learning Theory (COLT)*, 2025.

6. **Understanding Aggregations of Proper Learners in Multiclass Classification**
Julian Asilis, Mikael Møller Høgsgaard, Grigoris Velegkas.
In *Conference on Algorithmic Learning Theory (ALT)*, 2025.

5. **Proper Learnability and the Role of Unlabeled Data**
Julian Asilis, Siddartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
In *Conference on Algorithmic Learning Theory (ALT)*, 2025.

4. **Transductive Sample Complexities Are Compact**
Julian Asilis, Siddartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
In *Neural Information Processing Systems (NeurIPS)*, 2024.

3. **Open Problem: Can Local Regularization Learn All Multiclass Problems?**
Julian Asilis, Siddartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
In *Conference on Learning Theory (COLT)*, 2024. (Open problems track)

2. **Regularization and Optimal Multiclass Learning**
Julian Asilis, Siddartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
In *Conference on Learning Theory (COLT)*, 2024.

1. **Computable PAC Learning of Continuous Features**
Nathanael Ackerman, Julian Asilis, Jieqi Di, Cameron Freer, and Jean-Baptiste Tristan.
In *Logic in Computer Science (LICS)*, 2022.

## Skills

**Programming**: Python
**Languages**: English, Spanish